Business Recommendations

Term Deposit Customer Ranking

Executive decision. Approve controlled pilot planning using **unweighted Logistic Regression with all 12 valid pre-call predictors**. Retain HistGradientBoosting as a challenger. **Do not approve unrestricted production deployment yet.**

What the model does

The model scores every already eligible customer **before the call**, ranks scores from highest to lowest, and lets the call centre work down the list until capacity is exhausted. It is a prioritisation tool, not an eligibility, pricing, credit or consent decision system.

Historical evidence	Result
Dataset	40,000 calls; 2,896 subscriptions (7.24%)
Tuned shuffled holdout	HGB PR-AUC 0.2728 / ROC-AUC 0.7339; LR 0.2403 / 0.7097
Ordered expanding-window	LR PR-AUC 0.1126 / ROC-AUC 0.5663; HGB 0.0963 / 0.5516
LR at top 5%	1,528 calls; 228 subscriptions; ~123 expected at random; 1.854x lift

The ordered result matters most for the pilot decision because it tests later source-order blocks rather than randomly mixing historical records. Both models weaken materially, but LR is the more robust of the two.

Why the apparent accuracy is not the decision metric

With only 7.24% subscribers, a trivial majority rule is already 92.76% accurate. The business question is therefore whether the score orders subscribers toward the top of the call list. PR-AUC, ROC-AUC, ordered robustness and lift are more informative than raw accuracy.

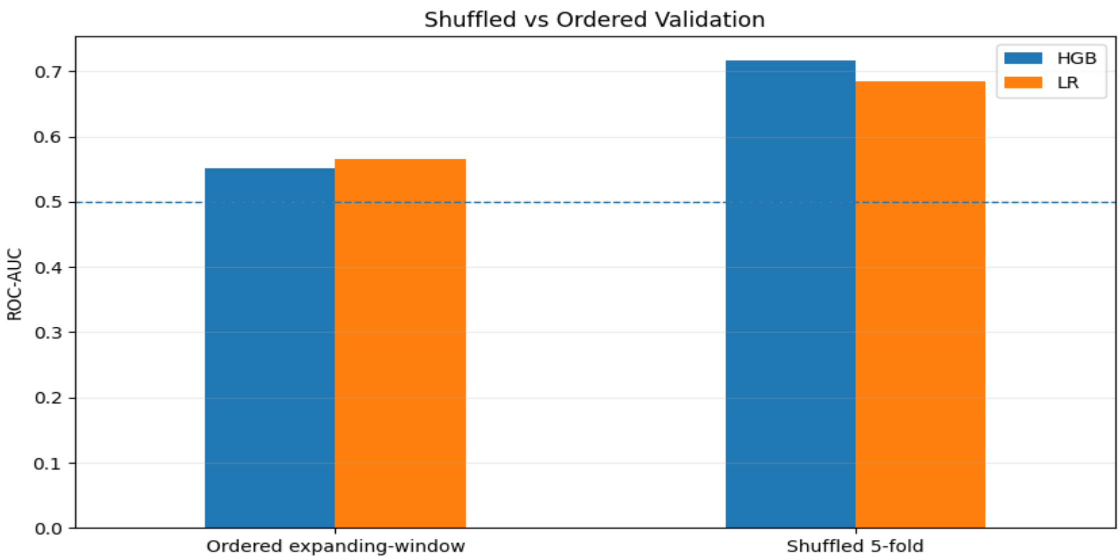
Important constraint. duration is only known after a call finishes. Its strong relationship with subscription is useful leakage evidence, but it cannot be used to decide whom to call.

1. Why Logistic Regression is the pilot champion

HistGradientBoosting ranks better when historical records are shuffled, but the advantage does not survive the stricter ordered test. LR also produces the stronger historical call-depth lift and is simpler to explain, monitor and challenge during a controlled pilot.

Validation	LR ROC-AUC	HGB ROC-AUC	LR PR-AUC	HGB PR-AUC
Shuffled 5-fold	0.6848	0.7175	0.2154	0.2453
Pseudo-profile sensitivity	0.6850	0.7167	0.2157	0.2481
Ordered expanding-window	0.5663	0.5516	0.1126	0.0963

Fig. 1: Shuffled vs Ordered Validation



How tuning should be interpreted

The **baseline pre-call and +duration versions use the same default hyperparameters within each model family**; only the available features change. Tuning is then performed separately for LR pre-call, LR + duration, HGB pre-call and HGB + duration, yielding four different tuned winners. Tuning itself changes pre-call ranking only modestly; the business choice is driven by robustness and lift rather than a claim that tuning solved instability.

Selected pre-call structures

Model	Pilot/challenger tuned pre-call structure
LR	L2; C=8.4834; unweighted
HGB	200 iterations; depth 8; learning rate 0.05; 15 leaves; L2=0; min leaf 20; unweighted; early stopping

2. What ranking can buy at limited call capacity

Lift asks a simple operational question: if the bank calls only the highest-ranked fraction of each eligible period, how concentrated are subscribers compared with random calling at the same call volume?

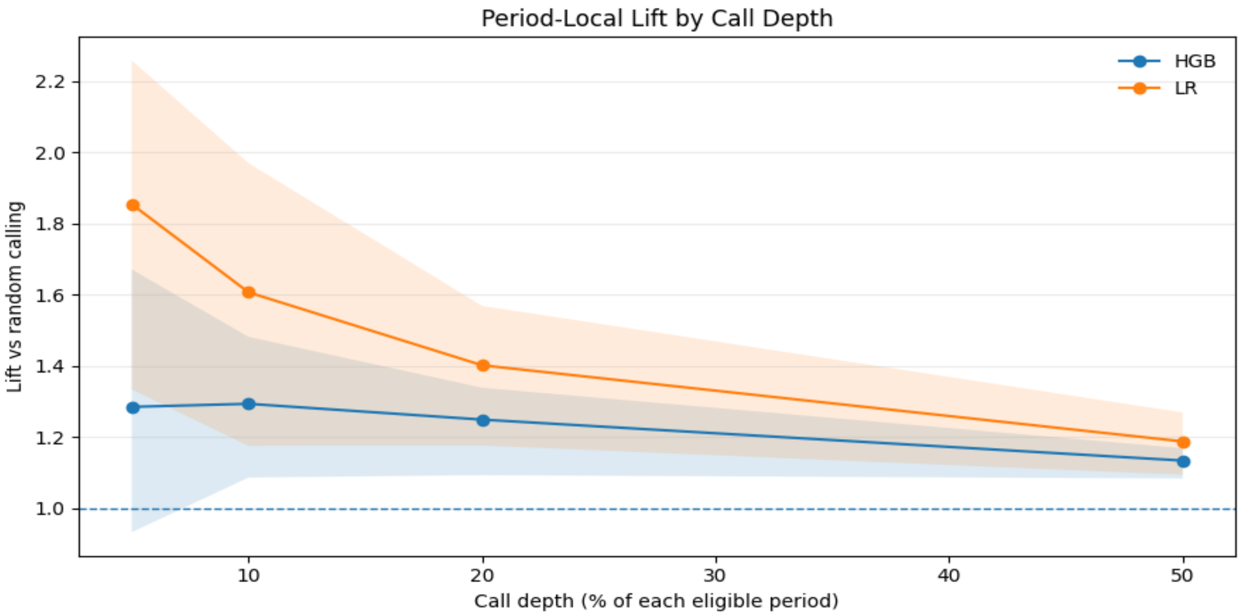


Fig. 2: Period-Local Lift by Call Depth

Call depth	LR calls	LR subscribers	Random expectation	LR lift
5%	1,528	228	122.99	1.8539x
10%	3,055	395	245.89	1.6064x
20%	6,108	689	491.63	1.4015x
50%	15,264	1,459	1,228.58	1.1875x

Do not turn these depths into quotas. 5%, 10%, 20% and 50% are retrospective sensitivity points. The bank should choose call depth from available capacity, expected net contribution, contact costs and risk controls.

Uncertainty and model comparison

LR has the strongest historical lift across the evaluated depths. At 5%, its period-level bootstrap confidence interval is approximately 1.33-2.26. HGB reaches 1.285x at 5%, but its interval includes 1.0, so the evidence for enrichment at that narrow depth is weaker. These are historical ranking results, not causal estimates of the incremental effect of calling.

3. Who should be called - and what not to infer

Operational rule: score every already eligible customer using the 12 pre-call variables, sort descending, and call from the top until the chosen capacity is reached.

Exploratory higher-conversion population

A descriptive population - contact=cellular and (job=retired or age >= 71 or balance >= Q3) - contains 6,774 customers, 778 subscribers and an 11.49% historical conversion rate. That is useful context, but it is **not** a validated demographic targeting rule and should not replace model ranking.

Guardrails for interpretation

Do not use duration. It is post-call and unavailable at ranking time.
Do not treat scores as exact probabilities. Ordered calibration diagnostics show non-trivial error and no sigmoid or isotonic recalibration was fitted.
Do not infer causality from feature importance. Variables such as month and contact channel help rank historical outcomes; they do not prove that changing those attributes would cause subscription.
Do not claim customer-level independence. The pseudo-profile grouped check uses age + job + marital + education + balance because no true customer identifier exists.

Primary data risk

contact=unknown occurs in 12,765 records (31.9%). Better contact-channel and CRM history are likely to improve ranking quality more than further retrospective model complexity.

Score reliability

Model	Ordered Brier	Ordered ECE	Interpretation
LR	0.0748	0.0444	Use as relative rank; prospectively recalibrate before probability economics
HGB	0.0742	0.0351	Same caveat; better calibration diagnostic does not outweigh weaker ordered ranking

4. Controlled pilot, economics and decision gate

The next step is a prospective equal-capacity experiment. The pilot should test whether ranking creates incremental business value under current campaign conditions, while explicitly monitoring customer and operational risk.

Arm	Purpose	Rule
LR-ranked	Primary treatment	Score eligible customers with final all-12 LR; call highest-ranked first
Comparator	Business benchmark	Equal-capacity BAU or random eligible-customer assignment
HGB challenger	Optional challenger	Equal capacity if operationally feasible; same outcome definitions

Primary outcome

Subscriptions per 1,000 assigned calls. Secondary monitoring should include contactability, complaints, opt-outs, repeat contacts, customer mix and agent time where available.

Stop / review triggers

Pause or review if customer outcomes deteriorate materially, operational data quality fails, model-ranked performance does not beat the equal-capacity comparator, or model performance changes materially across successive pilot windows.

Economics

The historical data do not contain the bank's actual call cost or net contribution per successful term deposit, so this report does not invent ROI. Once supplied, evaluate:

Incremental net value = incremental subscriptions x net contribution per subscription - incremental operating cost.
Probability-based customer-level expected value should wait for prospective recalibration.

Implementation priorities

Freeze eligibility and capacity before scoring; generate the LR-ranked list before calls begin; preserve comparator assignment and audit fields; record reliable timestamps, campaign IDs and contact outcomes; then evaluate lift and calibration prospectively before broader deployment.

FINAL DECISION: APPROVE FOR CONTROLLED PILOT PLANNING. DO NOT APPROVE FOR UNRESTRICTED PRODUCTION DEPLOYMENT.
Historical ranking evidence supports LR prioritisation, but the ordered-performance drop requires prospective confirmation.

Evidence source: all analytical metrics, tables and report figures are produced by the final executed notebooks 01-05 in the repository. No confusion matrices or report-only predictive analyses are used.